

Lecture 11 Tutorial — Answers

Q1. AI Lifecycle and Iteration

Correct order:

$$E \rightarrow A \rightarrow B \rightarrow C \rightarrow D$$

That is:

Formulate Problem $\rightarrow$ Collect Data $\rightarrow$ Feature Extraction $\rightarrow$ Machine Learning $\rightarrow$ System Deployed

If the deployed system does not work, the two explicitly suggested actions are:

(ii) Refine feature, (iii) New data

Q2. Concerns vs Practices

1-B, 2-C, 3-A, 4-D

1. Data privacy $\rightarrow$ **B. Anonymisation / federated learning idea**
2. Data collection bias $\rightarrow$ **C. Know your population**
3. False media $\rightarrow$ **A. Deepfake detection**
4. Power consumption of the system $\rightarrow$ **D. Algorithm simplification**

Q3. Federated Learning and Anonymisation

B. Transmitting data introduces a risk of data leaking; keeping data local reduces central exposure but may introduce inconsistency issues.

Q4. Data Collection Bias Mechanisms

The issues that may occur are:

- A. **A.** Inconsistent feature definition (or data collection)
- B. **B.** Inconsistent data quality
- C. **C.** Inconsistent feature distribution

The incorrect options are:

- D. **D.** Perfect matching of training and ground-truth distributions
- E. **E.** Automatic elimination of domain shifts

Q5. Dataset vs Ground Truth Distribution

1. **F.** The training dataset **never** has exactly the same probability distribution as the ground truth; it can only estimate it.
2. **T**

Q6. Domain Shift Responses

A-2, B-1, C-3

- A. Domain generalisation → **2.** Aim to perform well on unseen domains without direct target-domain tuning
- B. Domain adaptation → **1.** Learn from a source domain and adjust to a target domain with distribution shift
- C. Transfer learning → **3.** Reuse knowledge/representations from one task/domain to help learning in another

Q7. Algorithm Bias as Assumption Mismatch

A-3, B-2, C-1

- A. GMM → **3.** All clusters can be approximated by Gaussian distributions
- B. SVM → **2.** There exists a feature space that may classify data with a linear boundary
- C. Naive Bayes → **1.** Impacts of factors are independent of each other

Q8. Evaluation Bias and Metric Choice

- 1. Overall performance across all classes → **A. Accuracy**
- 2. Quality of positive predictions (minimise false positives) → **B. Precision**
- 3. Ability to identify the positive class (minimise false negatives) → **C. Recall**

Q9. Distance Measures

A-1, B-2, C-3

- A. Euclidean distance → **1.** Absolute distance
- B. Cosine distance → **2.** Directional distance
- C. Correlation distance → **3.** Temporal distance

Q10. Security Threats to AI Systems

- 1. Adding small, visually imperceptible noise to an image at test time to flip the prediction: **Perturbation**
- 2. Injecting a small number of maliciously crafted training samples into the dataset to degrade performance: **Data poisoning**
- 3. Adding barely audible noise to an audio clip at inference time to cause mis-transcription: **Perturbation**
- 4. Planting mislabeled samples into the training set so a trigger pattern causes a targeted misclassification: **Data poisoning**

Q11. Energy / Sustainability Calculation

$$\frac{60}{0.0005} = 120000$$

1.

120000

2. AlphaGo consumed approximately **120000** times the energy of the human reference.